users from having to worry about the exact sensors used. It aims to enable cloud-based 'Sensing-as-a-Service' and has developed cases for intelligent agriculture, smart production, urban crowd sensing, smart living and smart campuses.

Open source integration tools and platforms

DeviceHive offers a communication framework for connecting M2M devices to the IoT. It includes easy-to-use Web management software for networking, application security regulations and live monitoring devices. It also contains files on sample projects developed with DeviceHub, and has a simulation section where it provides visualisation of how DeviceHub works online.

DeviceHub.net could be considered the 'open source backbone of IoT'. It renders a cloud facility to monitor, track and control the devices, and stores the data collected from the Web page directly, and in real-time. It is mainly concerned with tracking health care information, the location of children and vehicles, monitoring weather, etc.

Open source toolkits

IoT Toolkit is a collection of libraries that enables communication with the latest IoT based environments and devices. This high-performance library collection is optimised for minimal memory consumption in RAM, ROM, high speed and versatility, on any device.

KinomaJS is a JavaScript based framework designed to create core embedded device applications. It has support for Linux, Mac OS X, Windows, Android and iOS.

Open source Node Flow editors

Node-RED is a Flow-based Internet of Things programming tool for connecting hardware devices, APIs and online services in an interesting and new way.

ThingBox is a set of ready-to-use software that is already installed and set up on an SD card. It's not a new automatic home box. It aims to help create new use cases that go much further than home automation.

Open source data visualisation

ThingSpeak is an IoT analytics service that allows you to aggregate, view and analyse live data streams from the cloud. It provides instant visualisation of data posted to ThingSpeak by your devices.

Freeboard is a free, open source dashboard project with optional hosted subscriptions that can be easily integrated and elegantly designed with a variety of data sources.

Open source home automation software

OpenHAB integrates different home automation systems, devices and technologies into a single solution. It is supplier- and hardware-neutral, and runs on any Java-activated system. One of its objectives is to enable users to add and combine new features to their devices.

Thing System comprises software and network protocols. It promises to find and bring together all the things in your home that are connected to the Internet so that you can control them. It supports a wide range of devices including Nest thermostats, Samsung smart air conditioners, Insteon LED bulbs, Roku, Google Chromecast, Pebble smart clocks, Goji smart locks and more. It is written in Node.js and can be connected to a Raspberry Pi.

Open source in-memory data grids (IMDG)

Hazelcast IMDG is often used as a database operating memory layer to improve application performance; distribute data across servers, clusters and geographies; ingest data at very high rates and manage large data sets.

Ehcache is a standards-based open source cache that improves performance, offloads your database and simplifies scalability. It is the most commonly used Java cache because it is robust, proven, fully functional and integrates with other popular libraries and frames.

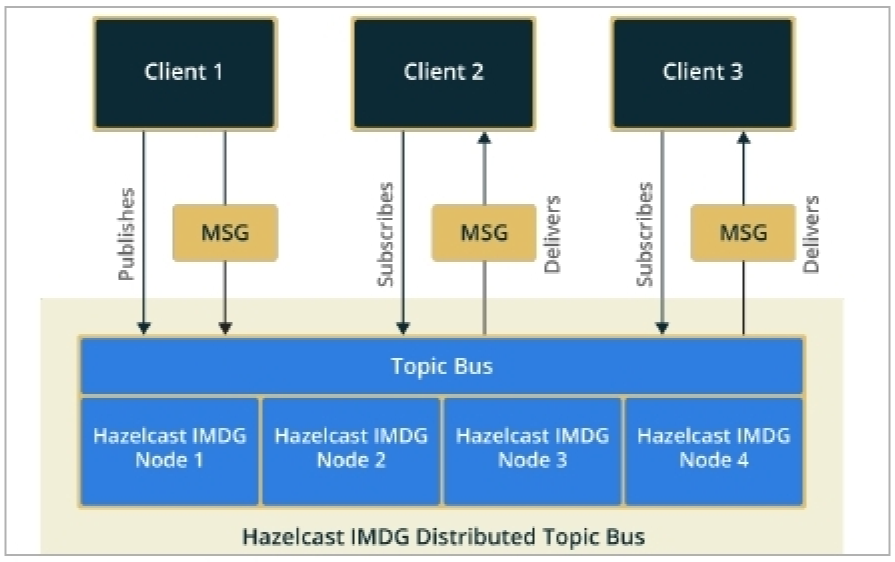

Figure 3: Hazelcast IMDG

The top ten IoT trends for 2019 and beyond

Gartner shared a write-up about the ten vital trends that will influence the Internet of Things (IoT), from 2019 through to 2023, at the Gartner Symposium/IT Expo in Barcelona, Spain in 2018. These are:

- New wireless networking technologies for IoT
- New IoT user experiences
- Social, legal and ethical IoT
- IoT governance
- Sensor innovation
- Artificial intelligence (AI)
- Infonomics and data broking
- The shift from the intelligent edge to the intelligent mesh
- Trusted hardware and operating systems
- Innovation on the chip END

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